Sean Grogan, PhD

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RESEARCH PROFILE

Operations researcher and applied mathematician working on optimization under uncertainty, logistics, and quantitative policy analysis with a strong geospatial component. My work applies stochastic and robust optimization, vehicle routing and location models, metaheuristics, and machine learning to public-sector problems: routing and pre-positioning UAVs for tornado damage assessment, and measuring the accessibility of voting and the representativeness of electoral data in Ontario. I have co-authored peer-reviewed work in both operations research and political science journals, and I bring practitioner experience leading the Business Intelligence function at a provincial electoral management body. Bilingual (English/French); eligible to work in Canada, the United States, and the European Economic Area.

RESEARCH AND TEACHING INTERESTS

Research	Stochastic and robust optimization; vehicle routing and facility location; metaheuristics; humanitarian logistics and disaster response; election administration and voter accessibility; spatial data analysis and GIS; record linkage and administrative data quality
Teaching	Operations research and optimization; applied statistics and data analysis; programming for engineers and scientists (Python, C++, SQL); GIS and spatial analysis; engineering mathematics; technical and scientific writing

EDUCATION

PhD, Industrial Engineering 2023
Polytechnique Montréal, Montréal, Québec
Thesis: *A data-driven approach to locating and routing unmanned aerial vehicles for disaster area reconnaissance*
Supervisors: Robert Pellerin and Michel Gamache

MSc, Computer Science and Operations Research 2016
Université de Montréal, Montréal, Québec
Thesis: *An adaptive neighborhood search algorithm for optimizing stochastic mining complexes*
Research conducted at COSMO – Stochastic Mine Planning Laboratory, McGill University Supervisors: Jacques Ferland, Roussos Dimitrakopoulos, Amina Lamghari

BEng, Mining and Materials Engineering (Co-op) 2013
McGill University, Montréal, Québec
Design project: Stochastic optimization of a mine using simulated annealing

ACADEMIC AND RESEARCH APPOINTMENTS

Postdoctoral Research Fellow Feb 2023 – Mar 2024
Laboratory of Elections and Democracy, Dept. of Political Science & Economics, Royal Military College of Canada *Kingston, ON*

- Research on demographic trends, spatial analysis, voting accessibility, and electoral logistics in partnership with government agencies (SSHRC Partnership Engage Grant).
- Built and validated a fuzzy record-linkage pipeline joining federal campaign-finance and census data ($\sim 10^6$ records), with match quality checked against a manually reviewed sample.
- Developed software to generate demographic profiles for custom geographies beyond standard Statistics Canada boundaries; maintained data infrastructure and research tools for the lab.

- Supervised and coordinated research assistants.

Research Engineer and Doctoral Candidate Sep 2016 – Apr 2023
Chaire de recherche Jarislowsky/SNC-Lavalin en gestion de projets internationaux, Polytechnique Montréal *Montréal, QC*

- Doctoral research in stochastic and robust optimization, vehicle routing, and location problems applied to disaster-response UAV logistics; three peer-reviewed articles and two conference papers.

Research Assistant Jan 2020 – Dec 2021
Laboratory of Elections and Democracy, Royal Military College of Canada *Kingston, ON*

- Analyzed the location and accessibility of polling stations; built demographic profiles from Statistics Canada and other open data. Co-authored an article in the *Canadian Journal of Political Science*.

Graduate Research Assistant 2014 – 2016
COSMO – Stochastic Mine Planning Laboratory, Dept. of Mining and Materials Engineering, McGill University *Montréal, QC*

- Research in long-range stochastic mine planning; adaptive metaheuristics for global optimization of mining complexes (up to 10% improvement in objective value).

PROFESSIONAL EXPERIENCE

Manager, Business Intelligence Apr 2024 – Present
Elections Ontario *Toronto, ON*

- First Business Intelligence manager at the agency; built the team from the ground up through five hiring campaigns (up to three direct reports).
- Lead analysis of the quality and representativeness of Ontario's Register of Electors ($\sim 10^7$ records), including comparisons with partner datasets of similar scale.
- Direct the team's execution of statutory reporting required under the *Election Act*, translating legal and regulatory requirements into data and reporting practice.
- Advise senior leadership on emerging election research and its implications for organizational strategy.

Long Range Mine Planner (Co-op) Apr – Sep 2013
Newmont Mining Corporation *Denver, CO, USA*

- Developed mixed-integer programming mine-scheduling tools (C++, CPLEX) in collaboration with the COSMO Lab.

Short Range Mine Planner (Co-op) Jan – Sep 2011
Syncrude Canada Ltd. *Fort McMurray, AB*

PUBLICATIONS

Peer-Reviewed Journal Articles

1. Garnett, H. A., & **Grogan, S.** (2025). The last-minute voter: Age and time of voting in Ontario, Canada. *Journal of Elections, Public Opinion and Parties*. doi:10.1080/17457289.2025.2568860
2. **Grogan, S.**, Gamache, M., & Pellerin, R. (2023). Using historical data to dynamically route post-disaster assessment unmanned aerial vehicles in the context of responding to tornadoes. *Applied Sciences*, 13(7), 4178. doi:10.3390/app13074178
3. **Grogan, S.**, Pellerin, R., Gamache, M., & Perrier, N. (2022). Location of disaster assessment UAVs using historical tornado data. *Geomatics, Natural Hazards and Risk*. doi:10.1080/19475705.2022.2115407

4. **Grogan, S.**, Pellerin, R., & Gamache, M. (2021). Using tornado-related weather data to route unmanned aerial vehicles to locate damage and victims. *OR Spectrum*. doi:10.1007/s00291-021-00640-1
5. Garnett, H. A., & **Grogan, S.** (2021). I came, I saw, I voted: Distance to polling locations and voter turnout in Ontario, Canada. *Canadian Journal of Political Science*. doi:10.1017/S0008423921000196

Conference Proceedings

6. **Grogan, S.**, Gamache, M., & Pellerin, R. (2019). Dynamically routing UAVs in the aftermath of a severe tornado. In *Proceedings of PROLOG 2019, Project and Logistics Conference*. Metz, France.
7. **Grogan, S.**, Gamache, M., & Pellerin, R. (2018). The use of unmanned aerial vehicles and drones in search and rescue operations: A survey. In *Proceedings of PROLOG 2018, Project and Logistics Conference*. Hull, UK.

Theses

8. **Grogan, S.** (2023). *A data-driven approach to locating and routing unmanned aerial vehicles for disaster area reconnaissance*. Doctoral dissertation, Polytechnique Montréal.
9. **Grogan, S.** (2016). *An adaptive neighborhood search algorithm for optimizing stochastic mining complexes*. Master's thesis, Université de Montréal.

Technical Reports

10. Garnett, H. A., & **Grogan, S.** (2023). *The Ontario voter experience: Understanding Ontario voters*. Internal report prepared for Elections Ontario.
11. **Grogan, S.** (2016). *Metaheuristic improvements to COSMO Mine Links*. Internal report prepared for Newmont Mining Corporation.

Data and Methods Contributions to Others' Research

12. Garnett, H. A. (2024). *Who gives? Who gets? Who wins? Who cares? Campaign finance in local races*. University of Toronto Press. — Processed Canadian census and federal campaign-finance data.
13. Garnett, H. A. (2024). Where do donors come from? Using census data to predict donations to Canadian federal election candidates. *Political Geography*. doi:10.1016/j.polgeo.2024.103077 — Processed Canadian census and federal campaign-finance data.
14. MacGregor, M. (2023). Public perceptions of Elections Ontario in the 2022 provincial election: A Twitter analysis. *Forum on Democratic Citizenship*. — Collected and pre-processed Twitter data.

Software

15. **Grogan, S.** (2019–present). *great-circle-calculator*: Great-circle distance, bearing, and midpoint computations [Python package]. PyPI.

Other Writing

16. **Grogan, S.** (2013, Sept. 19). Applying mathematical models to complex mining operations at Newmont. *McGill Reporter*.

GRANTS, FELLOWSHIPS, AND AWARDS

SSHRC Partnership Engage Grant (co-author of proposal) 2022
Social Sciences and Humanities Research Council of Canada; funded postdoctoral research at the Laboratory of Elections and Democracy

Jarislowsky Foundation Doctoral Fellowship 2016 – 2023
Polytechnique Montréal; renewed annually

Three Minute Thesis (3MT), Second Place 2018
Polytechnique Montréal; university-wide competition presenting doctoral research to non-specialist judges

NSERC graduate funding 2014 – 2016
Sustainable Mineral Resource Development and Optimization Under Uncertainty, COSMO Laboratory, McGill University

Entrance Scholarship 2008
McGill University

TEACHING AND INSTRUCTIONAL EXPERIENCE

Instructor, COSMO Mine Links Technical Training 2014 – 2016
COSMO Stochastic Mine Planning Laboratory, McGill University; Newmont Mining Corp. *Montréal, QC; Winnemucca, NV*

- Delivered multi-day lectures and one-on-one tutoring on stochastic mine-optimization software to industry partners at the annual COSMO Technical Days (2015, 2016).
- Led an on-site training program for Newmont Mining Corporation (2014), using mine-site visits to inform software improvements.

Technical Writing Tutor 2014 – 2023
Graduate researchers, McGill University and Polytechnique Montréal *Montréal, QC*

- Coached peers on clarity, concision, and structure in technical English; edited manuscripts to APA and IEEE style.

Head Coach, Lacrosse Program Sep 2011 – Dec 2013
Dawson College (CEGEP) *Montréal, QC*

- Designed season-long training curricula, practice plans, and game plans for college student-athletes; reported to college administration on program progress.

Instructor, National Officials Certification Program (Lacrosse) 2012 – 2014
Fédération de Crosse du Québec *Québec*

- Taught classroom theory and on-field practicum sessions to new and returning officials.

SUPERVISION AND MENTORSHIP

Business Intelligence Analysts, Elections Ontario 2024 – Present
Recruited, onboarded, trained, and evaluated up to three direct reports; designed an onboarding curriculum covering the Permanent Register of Electors, its technical implementation, and its legal context.

Research Assistants, Laboratory of Elections and Democracy, RMC 2023 – 2024
Supervised and coordinated research assistants on data and analysis projects.

Science Fair Mentor, Loyola High School, Montréal 2014 – 2024
Advised secondary students on research questions, experimental controls, and programming.

CONFERENCE PRESENTATIONS AND PARTICIPATION

Conference of Chief Electoral Officers, Victoria, BC 2023
Canadian Political Science Association (CPSA) Annual Conference, Western University, London, ON 2020
CSDC Conference – Democracy and Diversity, Centre for the Study of Democratic Citizenship, Montréal 2020
PROLOG 2019, IAE Metz School of Management, Université de Lorraine, Metz, France *(presenter; track coordinator)* 2019
Optimization Days (JOPT), HEC Montréal *(presenter)* 2018, 2019
INFORMS Annual Meeting, Phoenix, AZ *(presenter)* 2018
PROLOG 2018, University of Hull, UK *(presenter)* 2018
COSMO Stochastic Mine Planning Technical Days, McGill University 2015, 2016
CORS Annual Conference, Canadian Operational Research Society, Montréal 2015
Integrated Strategic Planning Seminar, Toronto 2014

ACADEMIC AND COMMUNITY SERVICE

Scientific Committee Member and Track Coordinator, Humanitarian Logistics and Disaster Response, PROLOG 2019 2019
Reviewed paper submissions and coordinated the track; IAE Metz School of Management, Université de Lorraine.

Pro Bono Analytics Volunteer (INFORMS), 10:12 Sports, Baltimore, MD 2021
Analyzed program and community-impact survey data and built dashboards for a youth nonprofit; documented the work so staff could maintain it.

Judge, Montreal Regional Science & Technology Fair (Réseau Technoscience) 2020 – 2024

Judge, Loyola High School Science Fair, Montréal 2014 – 2024

PROFESSIONAL DEVELOPMENT

Hiring and Selection Training, Brock University 2026 **Communications**, University of Toronto 2027 **Communications**, University of Toronto 2027

SKILLS

Methods	Mathematical modeling; mixed-integer, stochastic, and robust optimization; metaheuristics and hyper-heuristics; machine learning (clustering, classification, dimensionality reduction); spatial analysis; record linkage
Software	Python (scikit-learn, PyTorch); R; C/C++; SQL (SQL Server); CPLEX; Gurobi; QGIS; Power BI; LaTeX
Languages	English (native); French (professional working proficiency)
Work eligibility	Canada; United States; European Economic Area

PROFESSIONAL AFFILIATIONS

Association of Municipal Managers, Clerks and Treasurers of Ontario (AMCTO) Institute for Operations Research and the Management Sciences (INFORMS); Canadian Operational Research Society (CORS); Canadian Political Science Association (CPSA); Project Management Institute (PMI) and PMI-Montréal; CIRRELT; GERAD; Groupe de recherche en gestion et mondialisation de la technologie (GMT); Laboratory of Elections and Democracy (LED); Institute of Intergovernmental Relations, Queen's University; Montreal Operations Research Student Chapter (MORSC)

ONLINE

Scholar	Google Scholar profile	**ORCID**	orcid.org/0000-0002-1347-7874
GitHub	github.com/seangrogan	**LinkedIn**	linkedin.com/in/seangrogan
Research code	github.com/seangrogan-researchprojects	**Writing**	seangrogan.substack.com

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